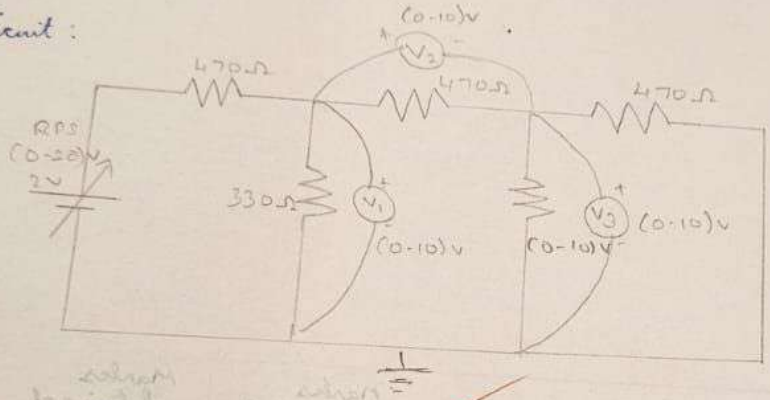
Circuit :



Exp. No: 03
DATE: 11/02/26

Aim:
To determine
both theoretic

Apparatus req

SL. NO	APPA
01	Reg
02	Mul
03	Res
04	Br

Circuit Diagr

Procedure:
1. Give con
2. Switch O
3. Note do
4. Vary th
5. Reduce
6. Verify

EXP. NO: 03
DATE: 11/02/26

Determination of voltage in circuit using Nodal analysis

Aim:
To determine the voltage in the circuit using nodal analysis both theoritically and practically for a given DC circuit

Apparatus required:

SL. NO	APPARATUS	SPECIFICATION	QUANTITY
01	Regulated Power Supply	(0-30V)	1
02	Multimeter	-	1
03	Resistors	470Ω, 330Ω	3, 2
04	Bread Board	-	1

~~Circuit Diagram:~~

Procedure:

1. Give connections as per circuit diagram
2. Switch ON and set voltage
3. Note down voltmeter and ammeter readings
4. Vary the RPS and note down the readings
5. Reduce the RPS to it's minimum value and switch OFF
6. Verify Kirchoff's law, and verify it theoratically

Tabular Columns:

Parameters	Theoritical	Practical
V_A	0.63	0.85
V_B	0.45	0.26
V_{AB}	0.19	0.59

Calculation:

Node A:

$$\frac{V_A - 2}{470} + \frac{V_A}{330} + \frac{V_A - V_B}{470} = 0$$

$$33(V_A - 2) + 47V_A + 33(V_A + V_B) = 0$$

$$113V_A - 33V_B = 66 \quad -(1)$$

Node B:

$$\frac{V_B - V_A}{470} + \frac{V_B}{330} + \frac{V_B}{470} = 0$$

$$33(V_B - V_A) + 47V_B + 33V_B = 0 \quad -(2)$$

Comparing (1) and (2)

$V_A = 0.64V$

$V_B = 0.45V$

$V_{AB} = 0.19V$

S.no	Description	Marks allotted	Marks obtained
01	Experimental Setup	20	20
02	observation / Reading	20	20
03	Calculation / Result	20	20
04	Viva	20	20
05	Record	20	20
	Total	100	100

100

10/2/26

Result:

Thus, the Nodal analysis is verified practically and theoritically.

The resultant voltages for ___ V supply are:

a) The Voltage V_1 is 0.64 V.

b) The voltage V_2 is 0.19 V.

c) the Voltage V_3 is 0.45 V.